

Antifouling and Bactericidal Zwitterionic Polymer Coatings with Synergistic Inhibitory and Killing Properties

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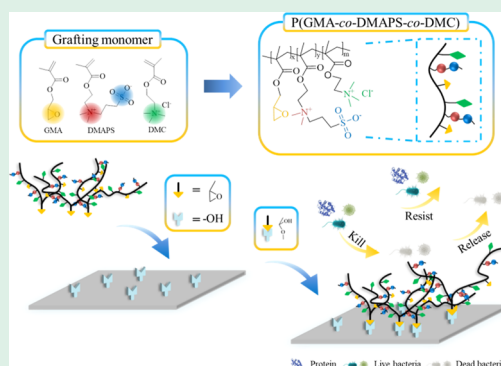
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ABSTRACT: Polymeric coatings that combine resistance to adhesion (“defending”) and killing (“attacking”) of biocontaminants were proposed to endow the surface with nonadhesive and bactericidal capabilities. In contrast, a zwitterionic copolymer P(GMA-co-DMAPS) with antifouling groups ([2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide, DMAPS) and a zwitterionic/cationic copolymer P(GMA-co-DMAPS-co-DMC) with bactericidal groups ([2-(methacryloyloxy)ethyl]trimethylammonium chloride, DMC) were synthesized, of which the latter exhibited synergistic inhibitory and killing properties. The distinct feed ratios of monomers were conducted, and the optimal molar ratio was obtained. The polymer coating was chemically bonded to the surface of substrates such as stainless steel via the addition–elimination reaction of epoxy groups and hydroxyl groups. Stainless steel surfaces grafted with G20D80-5DMC exhibited fine antifouling and bactericidal properties. The results showed that a highly efficient defending–attacking synergistic effect was achieved with antiprotein adsorption of more than 90% and bactericidal rates against *Staphylococcus aureus* and *Escherichia coli* of 98.6% and 96.6%, respectively. The polymers proposed in this study can effectively play an antifouling and antibacterial synergistic role and can be grafted onto substrates through a simple and effective method, showing attractive potential in marine, biomedical, and industrial applications.

KEYWORDS: zwitterionic polymers, coating, antifouling, bactericidal, synergistic effect



1. INTRODUCTION

Antibiofouling materials play an important role in numerous fields, such as ocean, pipeline, and medical fields. In the marine environment, not only are ships corroded by seawater but marine organisms also attach to the surface, causing biofouling. Traditional antifouling coatings achieve their antifouling properties by releasing antibiotics, silver, and quaternary ammonium salts (QAS) into the surrounding water environment, but harmful substances can be released from these coatings, causing harm to the ocean and marine life. Therefore, seeking environmentally friendly antifouling and bactericidal materials has become crucial to solving this problem. Currently, there are two main antibiofouling strategies using surface modification:^{1–3} one is “defending” of using biocompatible polymers as antifouling coatings to prevent protein or bacterial adhesion, among which poly(ethylene glycol) (PEG)^{4,5} and zwitterionic polymers⁶ are typical defensive materials that can prevent protein and bacterial adsorption, and the other is “attacking” of using bactericides or drugs to actively kill bacteria, involving surface functionalization with silver nanoparticles (AgNPs), cationic polymers, antimicrobial peptides, or decoration with QAS, graphene oxide,^{7–12} which

can show certain bactericidal properties through release killing or contact killing.¹³

However, the above-mentioned approaches have limitations for long-term biofouling resistance in complex biological environments. Defensive approaches are not capable of preventing bacterial deposition and colonization effectively,¹⁴ whereas the “attacking” approach is affected by swift accumulation of dead bacterial cells on the surface, conceals the surface functional groups, and provides platforms that are accessible for microbial attachment and growth.¹⁵ Therefore, considering two approaches comprehensively, that is, combining passive antifouling and active sterilization, to build a more ideal antifouling and bactericidal surface is undoubtedly a realistic choice.^{16,17}

The classic “defending” approach is to modify the materials’ surface chemistry by introducing PEG or OEG (oligo(ethylene

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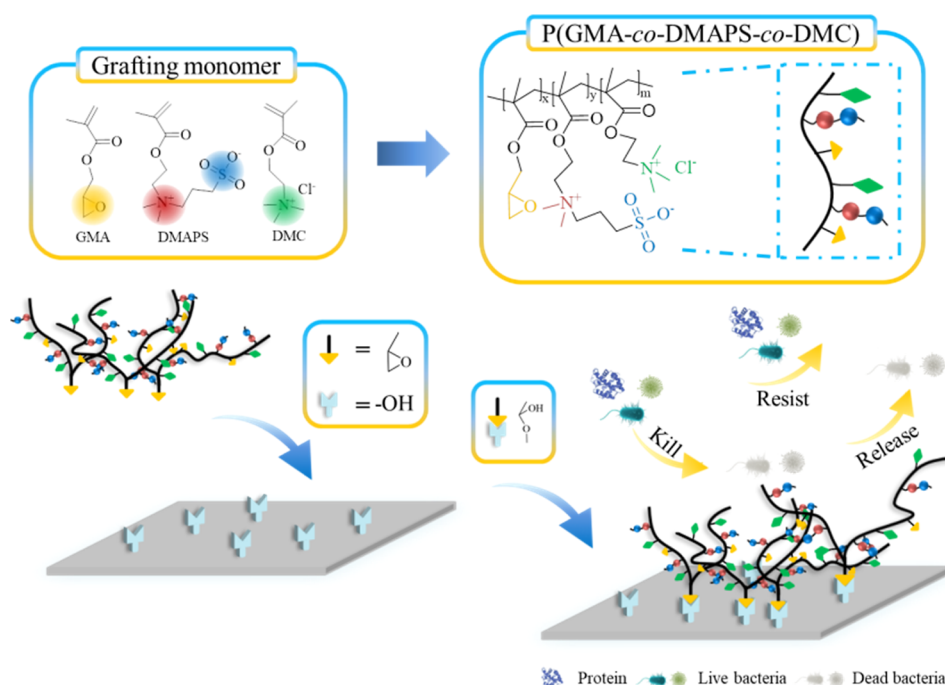
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Scheme 1. Surface Grafting Mechanism of P(GMA-*co*-DMAPS-*co*-DMC) Copolymers Grafted to Stainless Steel Surface via Chemical Bonds Formed between Epoxy Groups and Hydroxyl Groups



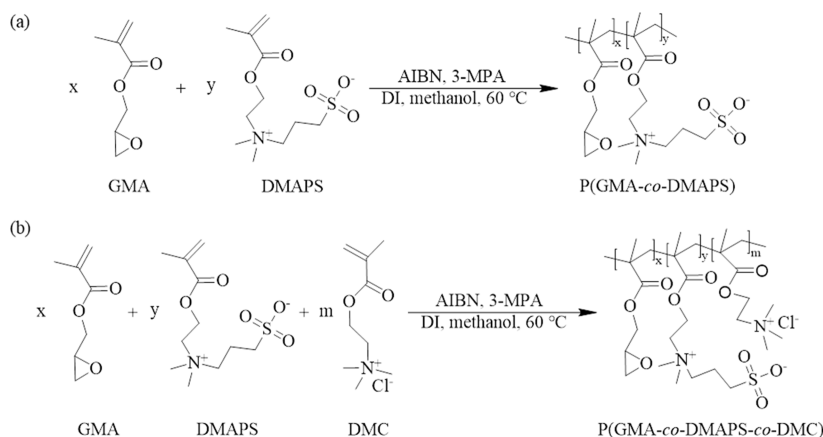
glycol)), which have the ability to form a hydration layer through hydrogen bonds with water molecules, thereby generating repulsive forces and nonadhesive properties.^{18,19} PEG is the most widely studied ligand with good resistance to nonspecific protein adsorption; however, it displays poor stability and can be easily oxidized in most biological environments,⁵ making it unsuitable for long-term utilization. A zwitterionic polymer with good resistance to a nonspecific protein and microbial adsorption is a promising antifouling material and an ideal alternative to PEG. Zwitterionic materials have a stronger ability to bind water than PEG materials because they can interact electrostatically to induce the formation of a stable hydration layer.^{18,20} The formation of the hydration layer and the steric hindrance properties of the hydrophilic groups can inhibit protein adsorption and cell and bacterial adsorption, and it has obvious biocompatibility and excellent antifouling properties. However, after modification with zwitterionic polymers, the material cannot inhibit bacterial growth and prevent bacteria from adhering to the surface of the material, which requires the introduction of some bactericidal substances. Chen²¹ successfully bound zwitterionic sulfopropylbetaine (ISB) and quaternary ammonium salt (IQAS) to prepare bactericidal and antifouling PVA nanofibers with low cytotoxicity. Jiang²² used SI-ATRP to graft the copolymer brush of sulfobetaine methacrylate (SBMA) and [2-(acryloyl)ethyl]trimethylammonium chloride (DAC) onto the surface of the cellulose film, improving the antifouling and bactericidal properties of the transparent cellulose film. The introduction of these bactericidal substances significantly improves the overall surface antifouling and bactericidal properties.

Although the combination of zwitterionic polymers and bactericidal substances shows good performance in antifouling and bactericidal properties, the poor stability is difficult to modify the multifunctional matrix.²³ Chen²⁴ demonstrated that zwitterionic polydopamine/poly(sulfonated methacrylate)

(PDA/PSB) has sufficient coating effect on the surface by physical adsorption anchoring, thereby achieving antiprotein and bactericidal properties, but the coated zwitterionic brushes or fragments have limited stability, resulting in limited antifouling surface coverage. Chou²⁵ prepared poly(2-oxoethyl methacrylate-methacrylic acid-*co*-sulfobetaine methacrylate) (OxMA-*co*-SBMA) and generated a Schiff base through nucleophilic addition reaction between the aldehyde group in the OxMA segment and the surface amino group. A stable and long-lasting antifouling surface was prepared using a fast and simple dip-coating technique. Jiang²⁶ used SI-ATRP to grow thin polymer brushes composed of SBMA and trifluoroethyl methacrylate on Si/SiO₂, which exerted long-term and stable antidiatom and deposition adsorption functions. It can be concluded whether through the simple reaction conditions of “grafting to” or the more flexible “grafting from” of grafted polymer brushes, stable chemical bonds can be formed on the surface to prepare efficient and durable modified surfaces.

Herein, a new environmentally benign and robust approach to integrate both “defending” and “attacking” antibiofouling strategies is proposed. We design a new environmentally friendly and effective method by combining DMAPS with a strong hydration ability, DMC with bactericidal ability, and glycidyl methacrylate (GMA) with strong adhesion. The epoxy functional groups in the polymers are bonded with the surface-anchored active groups, and the polymers are grafted onto the substrate surface in a “graft-to” manner to form “defending” and “attacking” functions (Scheme 1). The modified surface with both zwitterionic and quaternary groups, in which zwitterions play an antifouling role, while cationic groups interact with negatively charged bacterial cell membranes through an electrostatic interaction, can penetrate the cell membrane and lead to the loss of membrane potential and the death of bacteria, thus showing good antifouling and contact bactericidal properties. The obtained surface not only shows excellent antifouling performance in protein and bacterial

Scheme 2. Reaction Scheme of (a) GxDy and (b) GxDy-mDMC



adsorption but also possesses broad-spectrum antibacterial ability, which can effectively kill bacteria and inhibit the formation of a biofilm, thus achieving the dual effects of antifouling and antibacterial.

2. MATERIALS AND METHODS

2.1. Materials. [2-(Methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (DMAPS) was purchased from Shanghai Xingui Biotechnology Co., Ltd., China. GMA was purchased from Tianjin Heowns Biochem Technologie Co., Ltd., China. [2-(Methacryloyloxy)ethyl]trimethylammonium chloride (DMC) solution, triethylamine (TEA), Bis(carboxymethyl) trithiocarbonate (BCMT), and bovine serum albumin (BSA) were purchased from Shanghai Aladdin Reagent Co., Ltd., China. 2,2'-Azobis(2-methylpropionitrile) (AIBN) and 3-mercaptopropionic acid (3-MPA) were purchased from Shanghai Maclin Biochemical Technology Co., Ltd., China. *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) were purchased from the Beijing Microbiological Culture Collection Center. Nutrient Broth and Nutrient Agar Medium were purchased from Qingdao Hope Bio-Technology Co., Ltd., China. Methanol (MeOH) was purchased from Tianjin Fuyu Chemical Co., Ltd., China.

2.2. Synthesis of P(GMA-co-DMAPS). P(GMA-co-DMAPS) was prepared by free radical copolymerization (FRP) of zwitterionic DMAPS and the active monomer GMA (Scheme 2a). First, various amounts of GMA and SBMA monomers were separately added to methanol and water (volume ratio of 1:1 or 1:3). Then, the two solutions were mixed and stirred for 10 min to form a 1 mol/L solution. 1% equivalent of AIBN and 5% equivalent of 3-MPA were added, and then it was stirred appropriately to fully dissolve. Under nitrogen deoxygenation, the reaction was kept at 60 °C for 3 h and was terminated in an ice bath. After cooling for 15 min, there was a white precipitate occurred in the solution. The white product was redissolved in deionized water, reprecipitated in methanol, and then washed with methanol three times. Afterward, the product was dried to obtain a white crystalline powder (GxDy).

2.3. Synthesis of P(GMA-co-DMAPS-co-DMC). P(GMA-co-DMAPS-co-DMC) was prepared by FRP of zwitterionic DMAPS, reactive monomeric GMA, and functional monomeric DMC (Scheme 2b). DMC was predissolved in an aqueous solution of DMAPS, and the subsequent preparation steps were like those of GxDy. The resulting polymer was recorded as GxDy-mDMC.

2.4. Characterization of Copolymers. The molecular weights and their distributions (M_w/M_n) were determined by gel permeation chromatography (GPC, Waters-2695, USA). The eluent was an NaNO_3 aqueous solution of 0.1 mol/L at a rate of 1.0 mL/min at 23 °C. All the samples were prefiltered through 0.45 μm poly-(tetrafluoroethylene) filters for the experiments. The chemical structures of P(GMA-co-DMAPS) and P(GMA-co-DMAPS-co-DMC) copolymers were characterized by a nuclear magnetic

resonance (NMR) spectrometer (Bruker AVANCE III HD, USA). ^1H NMR spectra were conducted with a 500 MHz spectrometer using D_2O as the solvent. Fourier transform infrared (FT-IR, Bruker VERTEX 70, US) spectroscopy was used to detect the functional groups of monomers and copolymers with the detection background of KBr and a scanning wavenumber ranging from 4000 to 500 cm^{-1} . The number of scans was 32 times. The sample to be tested was tableted with KBr; that is, a small amount of sample was added to potassium bromide in the mortar, and the powder with the smallest possible particles was grinded. Finally, the ground powder was pressed into light transparent with an abrasive tool, and its transmittance was determined.

2.5. Preparation of Coatings. **2.5.1. Hydroxylation of Stainless Steel Sheets.** The stainless steel sheets (20 × 20 × 0.05 mm) were first cleaned with DI water and ethanol for 15 min under ultrasonication, followed by rinsing in the DI water. The cleaned stainless steel sheets were immersed in the newly formulated piranha solution and soaked for 30 min to enrich the surface with hydroxylation.²⁷ Finally, they were washed 3 times with DI water to neutral and stored in DI water for later use.

2.5.2. Surface Modification. After hydroxylation of the surface, the reactive epoxy functional groups of polymers are bonded with the surface-anchored active groups, and the copolymers were grafted onto the surface of stainless steel using the "grafting to" approach to prepare a polymer coating.

The treated substrates were immersed in aqueous solutions of P(GMA-co-DMAPS) at different concentrations, and different volumes of TEA were added to the coating solution. As is known, the epoxy group in GMA is induced by alkali to open ring and combine with the hydroxyl groups on the substrate to form a stable chemical bond. The reaction flask was placed under natural conditions for 24 h. After the grafting reaction, substrates containing S-GxDy or S-GxDy-mDMC were gently rinsed with DI water and dried for later use.

2.6. Surface Detection. The water contact angle (WCA) of the coating surface was measured by the seat drop method at 25 °C with the optical contact angle tester (SZ-CAMC33, Xuanzhun Co., Ltd., Shanghai, China). The samples were placed to be measured on the stage, and the camera focus was adjusted. Three μL of DI water droplets was dropped at three different sites on the sample surface to assess the surface wettability by measuring the average of the contact angles.

The physical structure of the modified surface was observed by a scanning electron microscope (SEM). To observe their surface topography more clearly, the sample on the stage was fixed and gold-plated in a vacuum environment for 60 s. Then, they were placed in the SEM chamber and vacuumed for observation at different magnifications. To further analyze and confirm the chemical structure and composition of the sample, X-ray photoelectron spectroscopy (XPS) (Shimadzu AXISSUPPA, Japan) was used to determine the surface elements and their contents.

2.7. Protein Adhesion. The antiprotein adhesion was determined by UV–visible spectroscopy. The protein solution displays a UV absorption peak at 275–280 nm. Within a certain concentration range, the absorbance of the protein solution at the maximum absorption wavelength is proportional to its concentration. The protein used in the experiment was the most common BSA. The absorbance of solutions with different concentrations at a wavelength of 280 nm was measured by a UV–visible spectrophotometer to draw a protein standard curve.

The samples were washed with phosphate-buffered saline (PBS, pH = 7.4) and immersed for 6 h to make them fully swell. After that, the samples were put into a BSA solution and cultured at 25 °C for 24 h. After incubation, the samples were gently rinsed with DI water and placed in a beaker with 2 mL of DI water for 20 min for sonication to fully desorb BSA on the samples. The absorbance of the desorption solution was measured using a UV–visible spectrophotometer at a wavelength of 280 nm to calculate BSA adsorption. The result was the average of three parallel samples.

2.8. Bacterial Attachment. *S. aureus* was selected by *S. aureus* and *E. coli* to characterize the antibacterial adhesion effect of the polymer coating. The bacteria were cultured in a fresh Nutrient Agar Medium slant and grown overnight at 37 °C until they grew to 10^8 CFU/mL for later use.

The fresh beveled culture of the test bacteria was washed with 0.9 wt % NaCl and diluted to 10^5 to 10^6 CFU/mL to prepare a bacterial suspension for later use. The samples were sterilized by soaking in 75 wt % alcohol, washed with PBS solution to remove residual alcohol, and dried until there was no obvious liquid on the surface. The experimental samples were soaked in 5 mL of bacterial solution with a concentration of 10^5 to 10^6 CFU/mL and cultured at 36 ± 1 °C for 72 h. After the incubation, the sample was soaked in 5 mL of PBS solution for ultrasonic elution for 10 min, 0.1 mL of the eluate was taken and diluted with 0.9 wt % of NaCl solution to $\sim 10^8$ CFU/mL, and 0.1 mL of sample solution was inoculated on the plate for viable bacteria count. At the same time, a blank sample that was the same as the sample was taken as the positive control group, and the same batch of 0.9 wt % NaCl solution and culture medium was taken as the negative control group. Experiments were repeated 3 times for each sample to calculate the average adsorption rate.

2.9. Bactericidal Test. *S. aureus* was selected by *S. aureus* and *E. coli* to characterize the antibacterial adhesion effect of the polymer coating. The bacteria were cultured in a fresh Nutrient Agar Medium slant and grown overnight at 37 °C until they grew to $\sim 10^8$ CFU/mL for later use. A sterile cotton swab was dipped into bacterial suspension, and it was spread evenly on a plate and the plate was covered. A hole was punched in the center of the dry plate, and 0.1 mL of polymer aqueous solution was added. The plate was covered and incubated at 36 ± 1 °C for 24 h, and then the results were observed.

The semioscillating flask method was used to quantitatively analyze the bactericidal properties. The samples were sterilized by soaking in 75 wt % alcohol, washed with PBS solution to remove residual alcohol, and dried until there was no obvious liquid. Experimental samples were immersed in 5 mL of bacterial solution of 10^5 to 10^6 CFU/mL and oscillated for 24 h at 36 ± 1 °C. 0.1 mL portion of cultured bacterial solution was taken and inoculated on a plate, and then viable bacteria were counted. Meanwhile, a blank sample was taken exactly as the positive control group, and the identical batch of 0.9 wt % NaCl solution and culture medium were taken as the negative control group. Experiments were repeated 3 times for each sample to calculate the average adsorption rate.

3. RESULTS AND DISCUSSION

3.1. Synthesis and Characterization of Copolymers.

In this study, zwitterionic polymers with synergistic inhibitory and killing effects were synthesized, breaking the traditional single-functional coating and integrating “defending” and “attacking”. The sulfonate betaine-based zwitterionic polymer with poor adhesion to the substrate material cannot form a

stable antiprotein adsorption surface.²⁸ Therefore, epoxy groups with high reactivity and easy ring-opening reaction to hydroxyl, carboxyl, and amino functional groups can be introduced,²⁹ so that the synthesized polymers can be grafted onto the surface of the substrate. The DMAPS segment was introduced for antifouling as well as the DMC segment for bactericidal properties and the GMA segment for fixing the copolymer on the surface. Furthermore, the DMAPS segments with strong hydration ability can create a favorable environment for the release of microorganisms killed by the DMC segments, thereby preventing the formation of biofilms that shield surface functional groups. In general, DMAPS segments can reduce protein adsorption on the surface and improve the antifouling level, and DMC segments can increase the bactericidal ability of the surface.

In contrast, DMAPS exhibits good solubility in an aqueous solution, while GMA is soluble in most common organic solvents. The choice of a suitable solvent is the key to obtaining the desired polymer. In this work, a mixed solution of water and methanol was used as a solvent for the polymerization reaction. GxDy with different molar ratios of GMA to DMAPS was synthesized via FRP. The molecular weights and dispersion coefficients of GxDy are shown in Table 1. The molecular weight of GxDy ranges from 8 to 20

Table 1. Molecular Weights and Dispersity Index of GxDy

sample ID	feed molar ratio of monomers	solvent ratio (v/v)	characterization of copolymers	
	DMAPS/GMA	methanol/DI	M_w /(kDa)	PDI
G20D80	20:80	1:1	17.6	1.87
G20D80	20:80	3:1	19.3	1.72
G40D60	40:60	1:1	10.8	1.53
G40D60	40:60	3:1	16.9	2.01
G60D40	60:40	1:1	8.6	1.92
G60D40	60:40	3:1	8.7	1.94
G80D20	80:20	1:1		
G80D20	80:20	3:1		

kDa, which decreases with a decrease of DMAPS dosage. The molar ratio of GMA to DMAPS was fixed at 20:80, the reaction solvent was a mixed solution of methanol and water with a volume ratio of 1:1, and the resultant polymer possessed a moderate molecular weight and a uniform distribution and good solubility. Under this condition, GxDy-mDMC with distinct DMC dosages was synthesized, and the molecular weight and their distribution are shown in Table 2. The molecular weights and their distributions of the obtained polymer tend to decrease with an increase in DMC dosage.

The functional groups in the copolymer were detected by FT-IR spectroscopy (Figure 1). FT-IR cannot accurately quantify the functional groups; a rough comparison can be made through the intensity of the absorption peaks. Since the zwitterionic polymer is hydrophilic, it easily absorbs water when exposed to air. The broad peak is hydroxyl (–OH) at 3462 cm^{-1} . The stretching vibration peaks are those of saturated methyl (–CH₂) and methylene groups (–CH) at 2972 cm^{-1} . Among the functional groups in G20D80, the main absorption peaks are the sulfonic acid group (–SO₃H) at 1032 and 1185 cm^{-1} , the quaternary amine (C–N⁺) at 1488 cm^{-1} , the ester group (COO–) at 1700 cm^{-1} , and the epoxy groups (–CH–O–CH₂–) of GMA at 929 cm^{-1} . It can be shown that

Table 2. Molecular Weight and Dispersity Index of GxDy-mDMC

sample ID	feed molar ratio of monomers		solvent ratio (v/v)		characterization of copolymers	
	DMAPS/GMA/DMC		DI/methanol		M_w /(kDa)	PDI
D80G20-3DMC	80:20:3		1:1		34.1	2.48
D80G20-5DMC	80:20:5		1:1		15.4	2.27
D80G20-10DMC	80:20:10		1:1		8.4	1.93

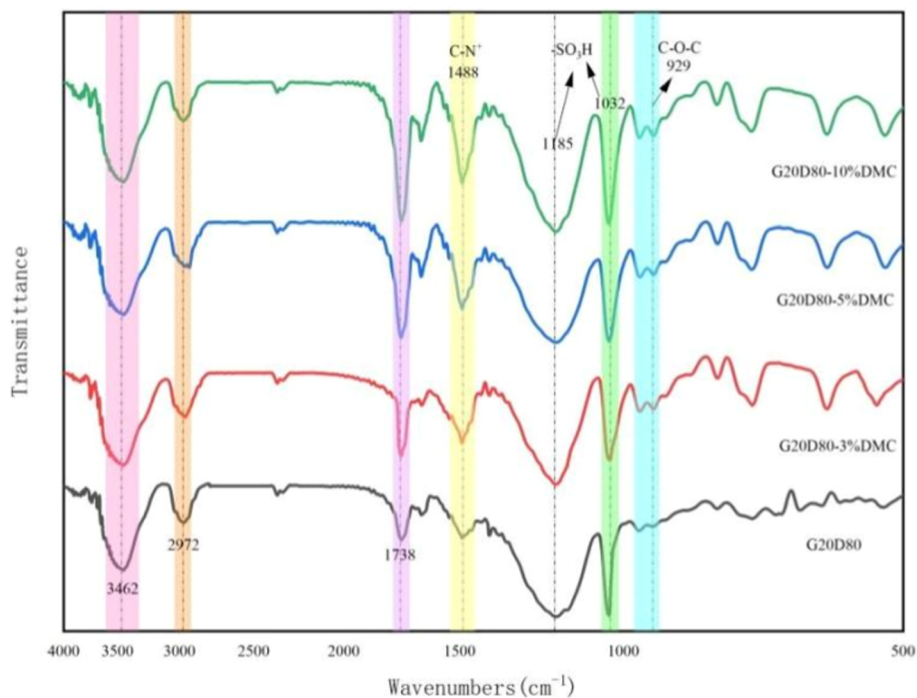
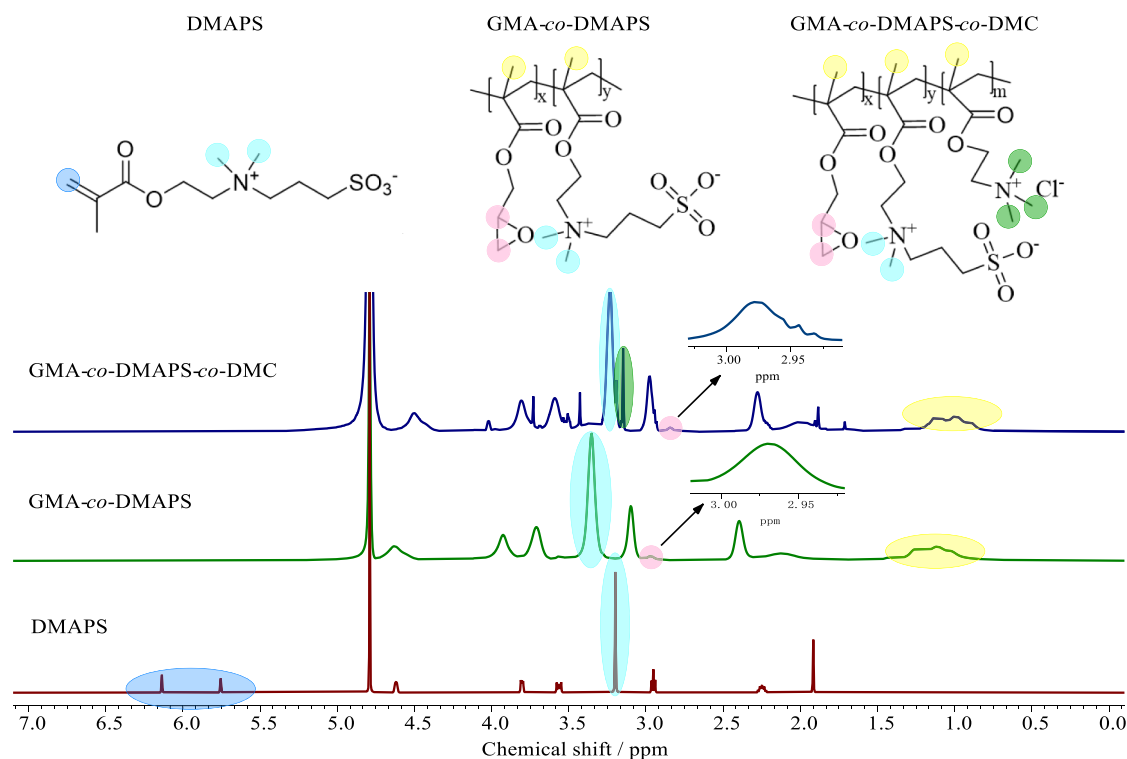


Figure 1. FT-IR spectra of GxDy and GxDy-mDMC and transient spectrum of GxDy and GxDy-mDMC.

Figure 2. ^1H NMR spectra of DMAPS, GxDy, and GxDy-mDMC and MX spectra of DMAPS, GxDy, and GxDy-mDMC.

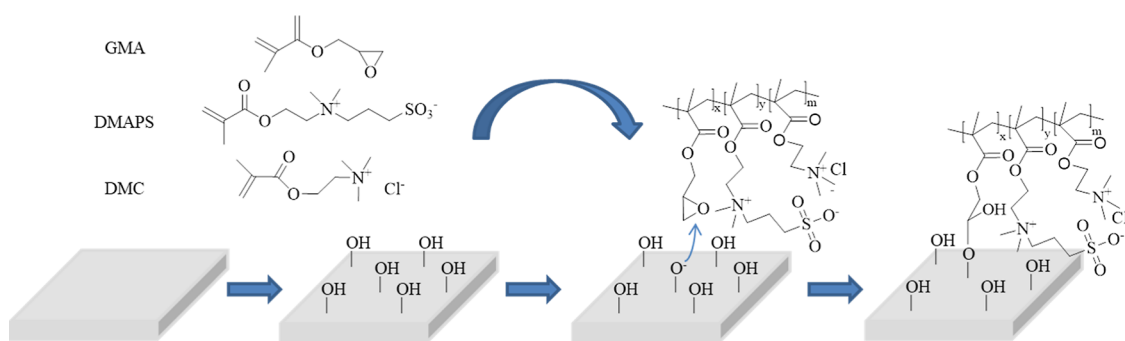


Figure 3. Mechanism of poly(GMA-*co*-DMAPS-*co*-DMC) grafted to stainless steel.

G20D80 has been successfully polymerized with GMA to obtain a zwitterionic copolymer with active epoxy groups.

P(GMA-*co*-DMAPS-*co*-DMC) has similar characteristic peaks to G20D80, with no obvious manifestation, but the absorbance of the stretching vibration peak of C–N⁺ of the positively charged group at 1488 cm^{−1} enhances, which is related to the fact that the introduced functional monomer DMC contains a quaternary ammonium cation. The absorbance of other relevant characteristic absorption peaks also increased accordingly.

To further verify the structural composition of the polymer, NMR was employed to characterize. Figure 2 shows the ¹H NMR spectra of G20D80 and G20D80-SDMC copolymers as well as the DMAPS monomer. The absorption peaks of hydrogen atoms directly connecting to C=C in DMAPS at 5.75 and 6.13 ppm disappear. This indicates that the unreacted monomers had been removed after purification treatment, and the obtained polymers were pure. In the ¹H NMR spectrum of G20D80, δ 2.97 ppm and δ 3.56 ppm signify the presence of –CH and –CH₂ of the epoxy group (–CH–O–CH₂–) in the GMA segment, while δ 3.35 ppm can be attributed the methyl group in the (CH₃)₂N⁺ segment of DMAPS. In the ¹H NMR spectrum of G20D80-SDMC, δ 3.24 ppm and δ 3.15 ppm correspond to the methyl group in the (CH₃)₂N⁺ of DMAPS and the DMC segment, respectively, while δ 2.84–3.43 ppm represents the methylene group of the epoxy group (–CH–O–CH₂–) in the GMA segment. Since the content of GMA in the zwitterionic copolymer is low and the epoxy functional groups are hydrophobic groups, they are mainly distributed inside the molecular chain; that is, they will be wrapped by the hydrophilic groups of DMAPS; therefore, the absorption peak is relatively weak.

In addition, the actual ratio of the two units in the copolymer molecular chain was obtained by calculating the characteristic peak area ratio of the copolymer. The actual molar ratio of GMA and DMAPS in G20D80 was 15.6:84.4, and the actual molar ratio of GMA, DMAPS, and DMC in G20D80-SDMC was 15.2:81.5:8.3, which is close to the actual feed ratio. In summary, polymers G20D80 and G20D80-SDMC were successfully prepared by FRP.

3.2. Preparation and Characterization of Copolymer-Grafted Surfaces. Since the surface of stainless steel is relatively stable, the surface needs to be activated to generate anchoring functional groups. In this work, piranha solution was used for surface treatment to generate hydroxyl groups,³⁰ and then the polymer was grafted onto the material surface through a base-catalyzed ring-opening reaction. The grafting process can be divided into two stages (Figure 3). In the first stage, active hydroxyl groups are generated on the original substrate

by piranha solution treatment. Due to the poor stability of hydroxyl groups, the treated samples must be quickly transferred to the coating solution. In the second stage, the substrate is immersed in an aqueous polymer solution with the addition of TEA. TEA as a catalyst could initiate a nucleophilic attack of hydroxyl groups on epoxy groups, and a chemical bond is formed by a ring-opening reaction. Most of the hydroxylated surfaces can be modified by this approach.

SEM was used to observe stainless steel modified with different polymers. The surface modified by G20D80-SDMC is shown in Figure 4. The coated surface is still flat from a

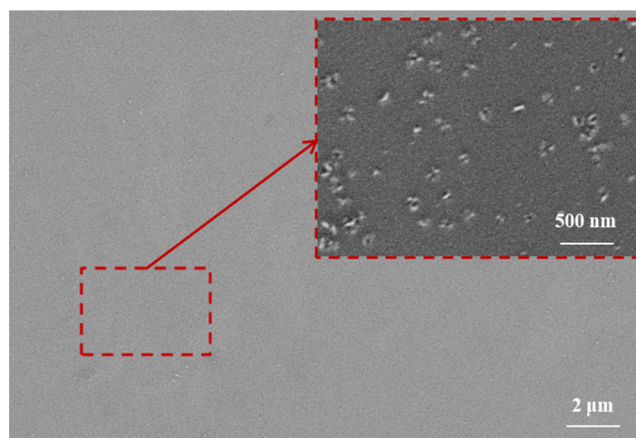


Figure 4. SEM images of stainless steel surfaces coated with G20D80-SDMC grafting.

macroscopic point of view. However, the magnification image shows obvious blocky polymer grafted on the substrate surface, forming an irregular rough surface. The polymer molecules are generally large, which makes the substrate surface rough due to spatial hindrance when they are grafted to the substrate surface.

To further confirm whether the polymer was successfully grafted onto the surface, XPS was used to analyze surface elements. From full spectrum scan of the untreated stainless steel surface (Figure 5), it can be clearly observed that the peaks of C 1s, N 1s and O 1s are located at 285.0 eV, 397.7, and 531.4 eV, respectively. In contrast, the peak at 1004.1 eV in the full spectra of S-G20D80 and S-G20D80-SDMC disappeared, and a new peak of S occurred at 166.2 eV, indicating that the polymers have been successfully grafted onto the stainless steel surface.

The surface was further scanned by a narrow spectrum of C, N, and S elements (Figure 6). From the C 1s narrow spectrum

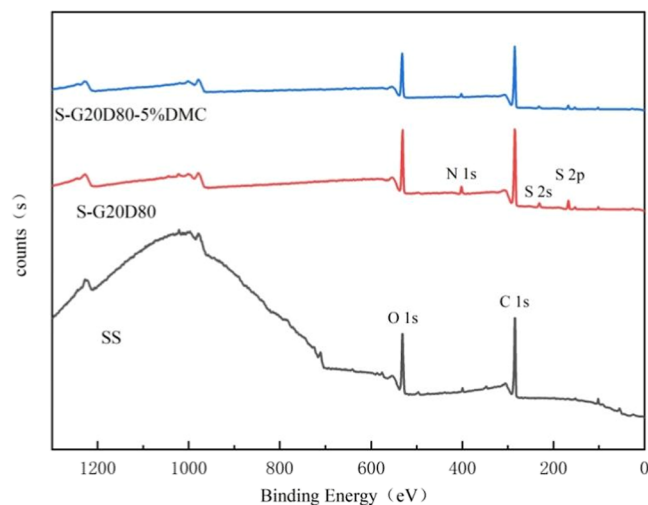


Figure 5. XPS spectra of stainless steel S-G20D80 and S-G20D80-5DMC.

curve, it can be seen that three peaks of 284.8, 285.6, and 288.5 eV were obtained after fitting on the original stainless steel surface, which represent the electron binding energies of C=O, C–O/C–N, and C–C bonds, respectively. The binding energies of C–O and C–N bonds are similar, so they are fitted into one peak, which is consistent with the surface structure of stainless steel. Three peaks also appeared in the surface fitting

of S-G20D80 and S-G20D80-5DMC, but the binding energy near 285 eV was significantly enhanced, which is due to the large number of C–O/C–N bonds in the polymer (Figure 6a–c).

From the N 1s narrow spectrum curve, it can be seen that there is a new peak at 402.2 eV on the grafted surface compared with the original stainless steel surface. The peak corresponds to the C–N⁺ bond, which is the characteristic peak of quaternary ammonium groups. Moreover, two new peaks appear in the narrow spectrum curve, corresponding to S 2p (Figure 6b,c), representing the 2p_{3/2} and 2p_{1/2} electron binding energies, respectively. The peak area ratio is 2.2:1, and the peak half-width height ratio is 1.1:1, close to the characteristic fitting peak of sulfur element, which is 2:1 and 1:1, respectively. The percentages of N and S elements observed on S-G20D80 and S-G20D80-5DMC are between 0.92 and 0.96. In summary, polymers were successfully grafted onto a stainless steel surface.

3.3. Protein Adhesion and Bactericidal Properties.

3.3.1. BSA Adsorption Test. Hydrophobic fragments may induce protein–surface interactions, leading to protein adsorption. Studies have found that zwitterionic polymers on the surface bind water via electrostatic interaction to form a hydration layer, thereby resisting protein adsorption and exhibiting good antiprotein adsorption properties.

The concentration of the polymer in the coating solution and the amount of catalyst TEA determine the grafting coverage of the polymer on the surface. Therefore, how to

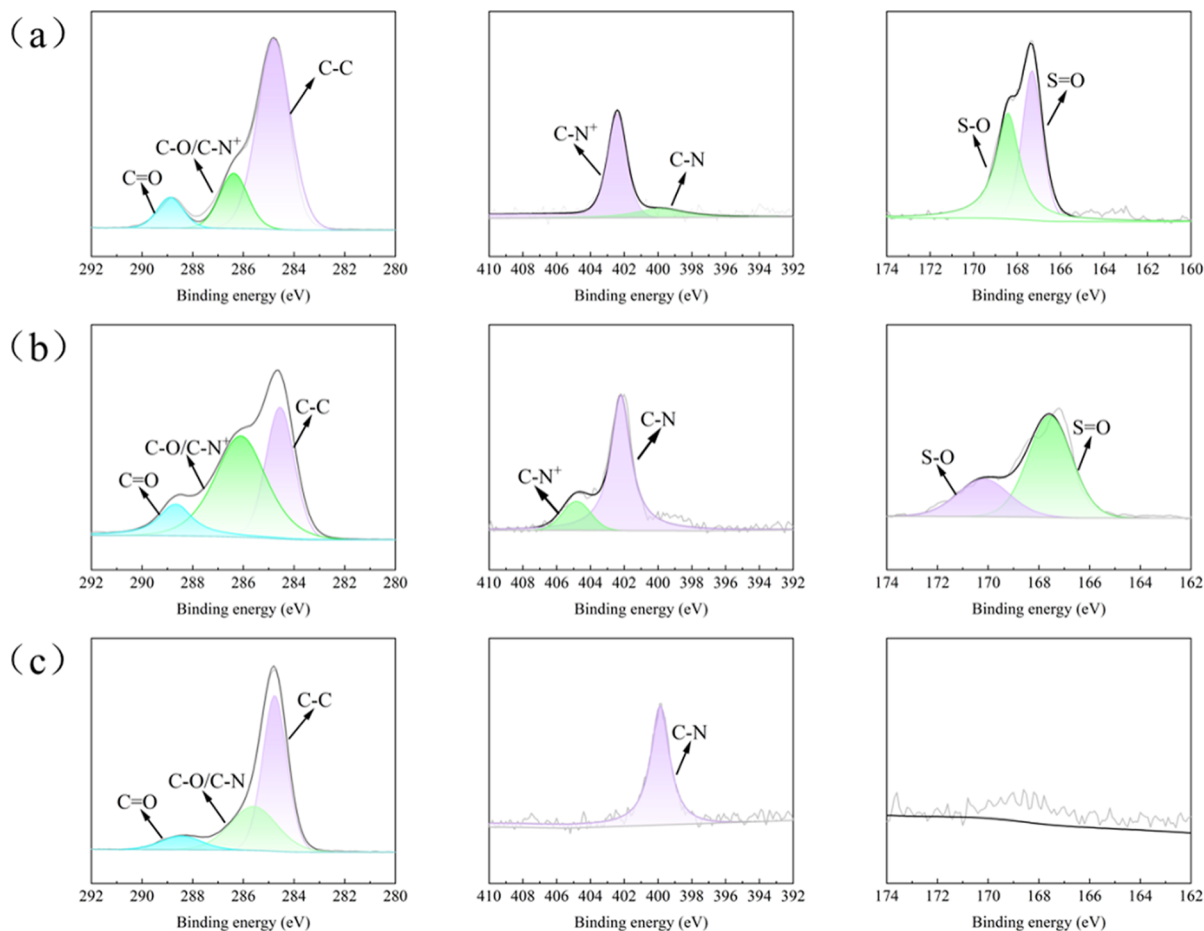


Figure 6. Curve fitting of the XPS narrow scan spectra of (a) S-G20D80-5DMC, (b) S-G20D80, and (c) stainless steel.

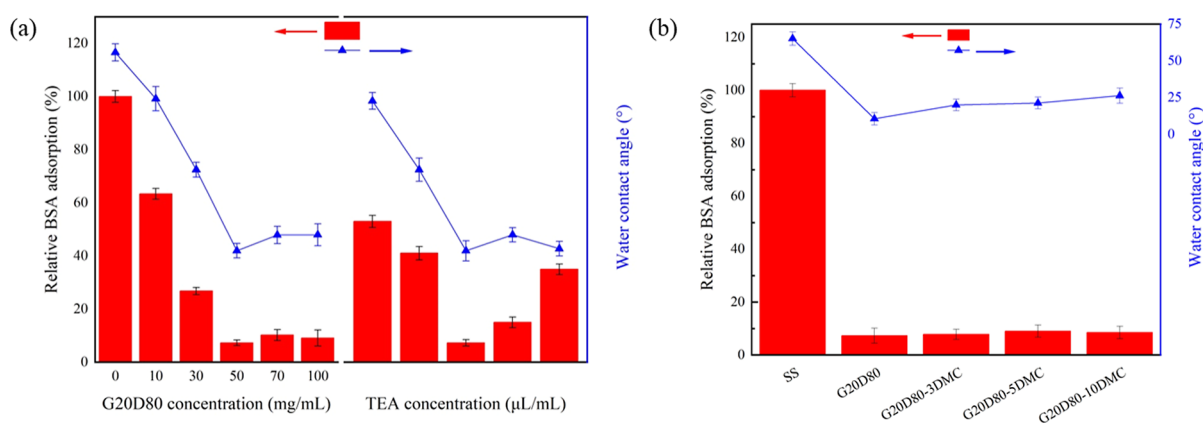


Figure 7. Changes in relative BSA adsorption and WCA of (a) S-G20D80 as a function of coating solution concentration and TEA dosage and (b) different molar ratios of DMC.

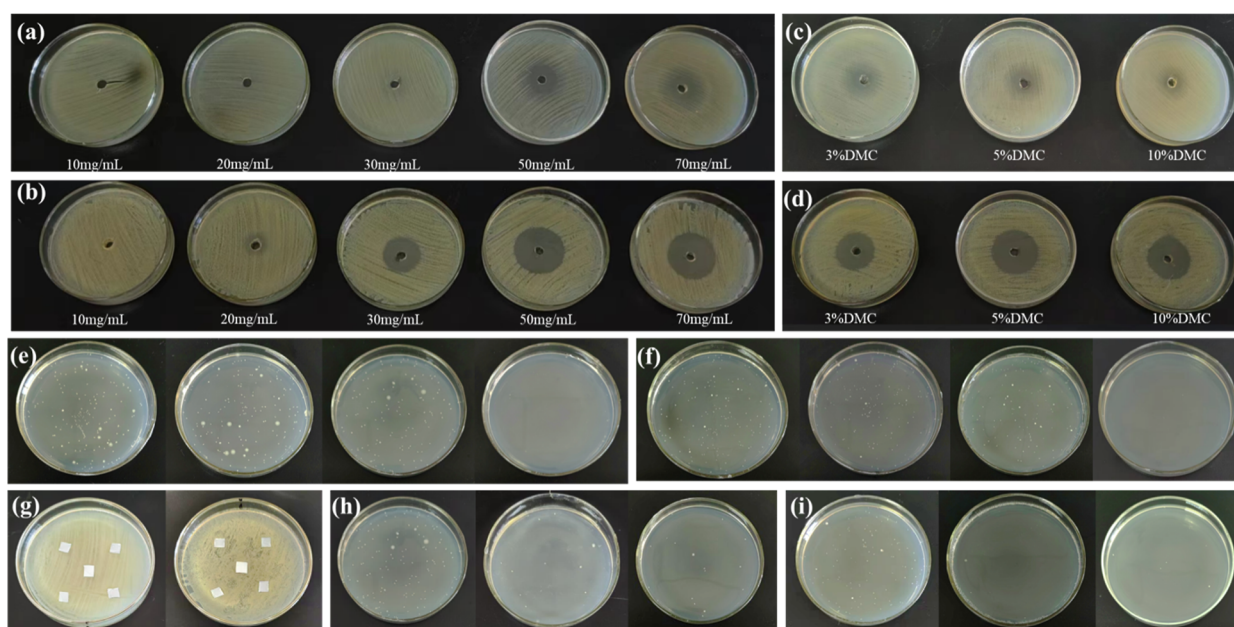


Figure 8. Bactericidal activity of polymers with different concentrations against (a) *E. coli* and (b) *S. aureus*; bactericidal properties of G20D80-mDMC with different DMC dosages on (c) *E. coli* and (d) *S. aureus*; bactericidal activity of the coating against (e) *E. coli* and (f) *S. aureus*; surface release bactericidal properties of the coating; and antiadhesion of different coatings against (h) *E. coli* and (i) *S. aureus*.

screen out the optimal coating conditions affects grafting rate on the surface and determines antifouling and bactericidal properties of coating. Herein, the hydrophilicity and antiprotein adsorption of the coatings with different polymer concentrations and different TEA dosages were studied (Figure 7a). Compared with the original surface of stainless steel, the contact angle exhibited a decreasing trend with an increase of polymer concentration and stabilized at 50 mg/mL. Correspondingly, the antiprotein adsorption capacity also showed a trend of decreasing and then stabilizing. The polymer cannot be effectively and uniformly grafted to the surface with lower concentration, resulting in poor hydrophilicity and protein resistance.

In addition, the amount of TEA is a key factor for epoxy ring opening and hydroxyl grafting. With the copolymer concentration of 50 mg/mL, even without TEA, the surface hydrophilicity and antiprotein adsorption can be improved, but its grafting efficiency is low due to the absence of TEA as a catalyst, which cannot fully exert the role of the polymer. As

the TEA concentration increases, the hydrophilicity of the polymer decreases and gradually tends to be stable, but its resistance to protein adsorption shows a trend of decreasing and then increasing. This is because the epoxy functional groups in the solution have basically been consumed when the TEA dosage reaches 100 $\mu\text{L/mL}$. However, the resistance to protein adsorption keeps increasing instead of decreasing when the TEA dosage continues to increase. The pH value of the polymer solution increases with an increase of TEA, resulting in the epoxy functional group to open prematurely, reducing the effective grafting rate on the hydroxylated surface and even inducing the zwitterionic polymer to hydrolyze. The optimal conditions for coating preparation can be determined with the polymer concentration of 50 mg/mL and the TEA concentration of 100 $\mu\text{L/mL}$. The best hydrophilicity and protein resistance can be achieved with a WCA of 10.63° and a protein resistance increasing ratio of 93%.

To further study whether the addition of bactericidal monomer DMC would affect the antiprotein performance of

the coating surface, we prepared DMC coatings with different dosages based on the optimal coating preparation conditions and measured WCA and antiprotein adsorption. Figure 7b displays antiprotein adsorption rate and WCA by different surfaces with the polymer concentration of 50 mg/mL and the TEA concentration of 100 μ L/mL. The antiprotein adsorption capacity of S-G20D80-mDMC is slightly lower than that of S-G20D80; overall, it is between the original stainless steel surface and S-G20D80. This is because negatively charged proteins will be attracted to the positive charges in DMC, resulting in an increase in the level of protein adsorption. This is also a common problem with surfaces modified by bactericidal materials.³¹ However, the surface hydrophilicity changes little after modification by G20D80-mDMC due to the formation of a tight hydration layer between the zwitterionic polymer and water molecules, which prevented contact of proteins. The introduction of zwitterion DMAPS enhances the hydrophilicity of the surface, and the interaction between water molecules and the coating surface is stronger, which strengthens the hydration capacity. When the protein approaches, the hydration layer acts as a barrier to prevent the protein from directly contacting the coating surface, thereby reducing the deposition of the protein on the surface. In addition, the polymer is electrically neutral, which reduces the interaction between the membrane surface and proteins with different charges, thereby alleviating the high adsorption of proteins on the membrane surface. In summary, this result shows that the addition of a small amount of DMC cannot make a significant effect on the antifouling performance; both G20D80- and G20D80-mDMC-modified surfaces display good antifouling properties.

3.3.2. Bacterial Attachment and Inactivation. The polymer concentration and the dosage of DMC play crucial roles in bactericidal properties of coating. We compared the antibacterial adhesion and bactericidal properties of different polymer concentrations and different DMC dosages. First, the minimum inhibitory concentration of polymer solution was tested, as shown in Figure 8. Figure 8a,b shows the bactericidal properties of different polymer concentrations on *E. coli* and *S. aureus*, respectively. Figure 8c,d shows the bactericidal properties of different DMC dosages on *E. coli* and *S. aureus*, respectively. As the polymer concentration increases, the bactericidal properties gradually become more significant. The minimum inhibitory concentration of DMC is 1 mg/mL. When the polymer concentration is lower than 20 mg/mL, the minimum inhibitory concentration cannot be reached, so there is no inhibitory zone. It can be clearly seen that as the concentration increases, the inhibition zone gradually becomes larger. When the polymer concentration reaches 50 mg/mL, the bactericidal performance tends to stabilize. The bactericidal properties achieve the best with the polymer concentration of 50 mg/mL and the DMC dosage of 5%. Therefore, the coating prepared by 50 mg/mL G20D80-5DMC was used to test the antiadhesion and bactericidal properties in the subsequent experiments.

The bactericidal performance and antibacterial adhesion performance of the G20D80-5DMC coating prepared under the optimal coating conditions were tested. For *S. aureus*, the adhesion amounts of S-G20D80-5DMC and S-G20D80 are not much different (Figure 8f). For *E. coli*, the adhesion amount of S-G20D80-5DMC is larger than that of S-G20D80 (Figure 8e). This observation may be due to the electrostatic attraction between the negatively charged *E. coli* cell and

positively charged quaternary ammonium part. Overall, the bacterial adhesion amount of coating surfaces decreased in the order of stainless steel > S-G20D80-5DMC > S-G20D80, indicating that both types of coatings have good antibacterial adhesion properties.

The bactericidal properties are listed in Figure 8g–i. The adhesion amount of live *E. coli* on S-G20D80 and S-G20D80-5DMC decreased from 90.35% (the amount of live bacteria on the original stainless steel surface) to 70.15% and 3.14%, respectively, and the number of live *S. aureus* decreased from 90.97% (the amount of live bacteria on the original stainless steel surface) to 69.28% and 1.20%, respectively. The positively charged quaternary ammonium groups in DMC can form cations with the cell wall, giving rise to the bacterial cell wall membrane to rupture, thereby endowing the S-G20D80-5DMC surface excellent bactericidal properties.

To verify whether the prepared coating is a release-type bactericidal material, the inhibition zone method was utilized for the evaluation in this experiment. The solid inhibition zone experiment was carried out with a stainless steel sheet modified with G20D80-5DMC. It can be observed that no inhibition zone appeared, indicating that its sterilization method is contact sterilization rather than release sterilization,³² which is more environmentally friendly. In general, S-G20D80-5DMC displays excellent and environmentally friendly bactericidal performance; however, S-G20D80 shows poor bactericidal performance with strong antiadhesion performance. The zwitterionic polymer still possesses the zwitterionic characteristics after being grafted to the substrate surface through epoxy ring-opening. It combines with water molecules and forms a hydration layer on the surface through an electrostatic interaction, which reduces contact between bacteria and surface and consequently exhibits good antibacterial adhesion.

3.3.3. Biocompatibility and Stability. On the one hand, DMAPS is a zwitterionic monomer used for the synthesis of poly(sulfobetaine), which is an electrically neutral polymer that contains both cationic and anionic groups in one monomer unit and has interesting properties such as electrolyte resistance, biocompatibility, and blood compatibility in solution. These properties make poly(sulfobetaine) useful for many biomedical applications such as drug delivery and drug formulations. Hoare et al. reported the test results of in vitro cytotoxicity and in vivo inhibition of PDMAPS, demonstrating good biocompatibility.³³ On the other hand, DMC is a cationic monomer that can be homopolymerized or copolymerized with other monomers to produce cationic polymers. Zhang et al. found that this bactericidal functional monomer is devoid of cytotoxicity and is applicable in biomedical area.³⁴ The copolymerization of DMAPS and DMC as well as GMA can suggest that the resulting GxDy-mDMC can possess favorable biocompatibility with extremely low cytotoxicity. Further biocompatibility studies including cytotoxicity assays are undergoing. In addition, although the polymeric coatings displayed ideal short-term stability, their long-term stability is still the focus of attention in this field, which will be conducted to evaluate their durability in complex environments such as marine or biomedical applications over extended periods.

4. CONCLUSIONS

We demonstrated a series of copolymers of P(GMA-co-DMAPS) and P(GMA-co-DMAPS-co-DMC) for multifunctional surface grafting. These copolymers exhibit universal antibiofouling properties, and P(GMA-co-DMAPS-co-DMC)

simultaneously possesses both antibiofouling and bactericidal capabilities. P(GMA-co-DMAPS) and P(GMA-co-DMAPS-co-DMC) copolymers are covalently grafted onto the hydroxylated stainless steel surface via ring-opening efficiency regulated by TEA. TEA as an alkaline catalyst promoted the ring-opening of epoxy groups and enhanced the grafting rate of zwitterionic copolymers. The polymers were screened by adjusting the feed ratio of monomers to explore optimal synthesis conditions. The polymer grafting rate was regulated by polymer concentration and TEA dosage, and the optimal preparation conditions were found based on surface hydrophilicity and antiadhesion. The surface modified by the copolymer G20D80-SDMC showed a 90% increase in antiprotein and antibacterial adhesion of the polymer coating compared with the original stainless steel surface, and the sterilization rates against *S. aureus* and *E. coli* reached 98.6% and 96.6%, respectively. Compared with S-G20D80, it has a broad-spectrum bactericidal property while retaining good antiprotein adsorption properties, which indicates that a zwitterion-modified surface with synergistic inhibitory and killing properties has been successfully prepared. There are both “defense” and “attack” properties on the surface, and its attack adopts contact killing, which can reduce the harm to the natural environment and be more environmentally friendly. A simple and universal method to construct a coating surface by covalently linking quaternary ammonium and zwitterionic polymer has been provided. Such excellent antifouling and bactericidal properties have great potential in the prevention and treatment of biofouling.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

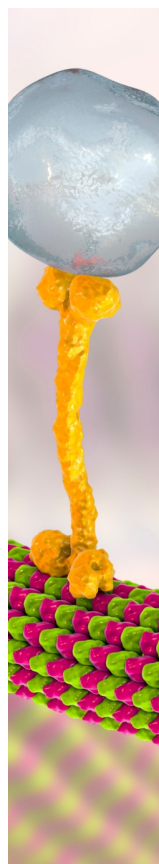
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